

n	proof	n	proof	n	proof	n	proof	n	proof
2	(5.7)	22	$\rightarrow 64$	42	$= 2$	62	$= 2$	82	$= 4$
3	(5.7)	23	(5.7)	43	$\rightarrow 64$	63	$\rightarrow 32$	83	(5.7)
4	(6.4)	24	$= 2$	44	$= 2$	64		84	$= 2$
5	(5.7)	25	(6.4)	45	(5.7)	65	$\rightarrow 9$	85	$\rightarrow 64$
6	$= 2$	26	$= 2$	46	$= 16$	66	$= 2$	86	$= 2$
7	$\rightarrow 4$	27	(8.10)	47	(5.7)	67	$\rightarrow 4$	87	$= 3$
8	(7.10)	28	$= 4$	48	$= 2$	68	$= 2$	88	$= 4$
9	(6.4)	29	$\rightarrow 8$	49	(6.4)	69	(5.7)	89	$\rightarrow 9$
10	$= 4$	30	$= 2$	50	$= 8$	70	$= 4$	90	$= 2$
11	(5.7)	31	$\rightarrow 16$	51	(5.7)	71	$\rightarrow 8$	91	$\rightarrow 16$
12	$= 2$	32	(8.10)	52	$= 4$	72	$= 2$	92	$= 8$
13	$\rightarrow 4$	33	$\rightarrow 9$	53	$\rightarrow 27$	73	$\rightarrow 4, 9, 25$	93	(5.7)
14	$= 2$	34	$= 4$	54	$= 2$	74	$= 2$	94	$\rightarrow 1024$
15	$\rightarrow 8$	35	$= 3$	55	$\rightarrow 4$	75	(5.7)	95	$= 3$
16	(8.10)	36	$= 8$	56	$= 2$	76	$= 16$	96	$= 2$
17	$\rightarrow 9$	37	$\rightarrow 4$	57	$\rightarrow 8, 9$	77	(5.7)	97	$\rightarrow 4, 9, 25, 49$
18	$= 2$	38	$= 2$	58	$= 4$	78	$= 8$	98	$= 2$
19	$\rightarrow 4$	39	$= 3$	59	(5.7)	79	$\rightarrow 4, 27$	99	$\rightarrow 8$
20	$= 2$	40	$= 4$	60	$= 2$	80	$= 2$	100	$= 4$
21	(5.7)	41	$\rightarrow 9$	61	$\rightarrow 16$	81		101	(5.7)

TABLE 1. Equational proofs for $2 \leq n \leq 101$ and their reductions

Theorem F (Theorems 7.5 and 7.9). *Let p be a prime number and $k \geq 1$. Assume that for all $1 \leq m < k$ there is an equational proof of the constructive Wedderburn Theorem W_{p,k,Tp^m} . Then there is an equational proof that every p^k -ring with $p = 0$ is commutative. The same conclusion holds when there are equational proofs of all $W_{p,k,f}$ where f is of degree $< k$.*

Thus, the problem reduces to finding proofs of certain constructive Wedderburn Theorems. We do not succeed in general, but the following special case already covers many cases.

Theorem G (Theorem 8.10). *Let p be a prime number and $k \geq 1$. If $\gcd(k, p^k - 1) = 1$, then there is an equational proof that every p^k -ring with $p = 0$ is commutative.*

Table 1 summarizes our results for $2 \leq n \leq 101$. Writing $\rightarrow m_1, m_2, \dots$ means that commutativity can be reduced to the classes of m_1 -rings, m_2 -rings, etc. (by Theorems 3.14, 4.1 and 5.5). Writing $= m$ means that the classes of n -rings and m -rings coincide. An empty cell means that the equational proof is not complete yet. We plan to fill these gaps in future work.

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2. Preliminaries

Following Jacobson [Ja12, Section 2.17], rings are assumed to be unital, and we explicitly speak of rings without a required unit when necessary, which we call *rngs*. See also [Po14]. Of course, rings are not assumed to be commutative unless otherwise stated. The set of prime numbers is denoted by $\mathbb{P}$.

We need to briefly formalize the notion of an equational proof in order to clarify what is allowed in our proofs and what not. For more details, we refer to [Bi35] and [Ta79].

Definition 2.1. If $\mathcal{T}$ is an algebraic theory and σ, τ are terms in $\mathcal{T}$, by an *equational proof* of $\sigma = \tau$ we mean a deduction $\mathcal{T} \vdash \sigma = \tau$ based on the following rules:

- (1) If $\sigma = \tau$ is an axiom of $\mathcal{T}$, then $\mathcal{T} \vdash \sigma = \tau$.
- (2) We have $\mathcal{T} \vdash \sigma = \sigma$.
- (3) If $\mathcal{T} \vdash \sigma = \tau$, then $\mathcal{T} \vdash \tau = \sigma$.
- (4) If $\mathcal{T} \vdash \sigma = \tau$ and $\mathcal{T} \vdash \tau = \rho$, then $\mathcal{T} \vdash \sigma = \rho$.
- (5) If ρ is a term of arity n in $\mathcal{T}$ and $\sigma_1, \dots, \sigma_n, \tau_1, \dots, \tau_n$ are terms in $\mathcal{T}$ with $\mathcal{T} \vdash \sigma_i = \tau_i$ for all $i = 1, \dots, n$, then $\mathcal{T} \vdash \rho(\sigma_1, \dots, \sigma_n) = \rho(\tau_1, \dots, \tau_n)$.
- (6) If σ, τ are terms of arity n in $\mathcal{T}$ with $\mathcal{T} \vdash \sigma = \tau$, then $\mathcal{T} \vdash \sigma(\rho_1, \dots, \rho_n) = \tau(\rho_1, \dots, \rho_n)$ for all terms $\rho_1, \dots, \rho_n$ in $\mathcal{T}$.

Rule (1) is self-explanatory. Rules (2),(3),(4) state that equational equality is an equivalence relation. Rule (5) means that we may replace equationally equal terms in expressions. Rule (6) allows substitutions of variables with more complex expressions. In practice, we will just write $\sigma = \tau$ instead of $\mathcal{T} \vdash \sigma = \tau$. Notice that there are no quantors and no negation.

Example 2.2. There is an equational proof that every group G with $g^2 = 1$ for all $g \in G$ is commutative. More precisely, in the theory of groups with the additional axiom $g^2 = 1$, we derive $gh = hg$ as follows: We have $(gh)^2 = 1$ by rule (6), and $1 = 1 \cdot 1 = g^2 \cdot h^2$ by rule (5). (Of course, we have also applied the other, more trivial rules.) Hence, $(gh)^2 = g^2 h^2$, which means $ghgh = gg hh$. Multiplying with g^{-1} on the left and with h^{-1} on the right, that is, plugging this equation into the term $g^{-1} x h^{-1}$ and using rule (5) again, shows $hg = gh$.

Remark 2.3. Notice that an equational proof does not talk about elements of a set. It is a purely *syntactical* deduction. Even when we write “For all elements $x, y, \dots$ ” (for the sake of readability of the proof), these are actually just variables. Alternatively, we may interpret an equational proof as a classical proof where every statement has the form “For all elements $\dots$, the equation $\dots$ holds”, and only a few basic deduction rules are allowed, namely the ones in Definition 2.1.

In the case of the algebraic theory of rings, we are not allowed to use ideals, subrings, quotient rings, direct products of rings, let alone more advanced techniques from commutative or non-commutative algebra. The LEM is not allowed either, so a commutativity proof by contradiction (“If the ring is not commutative, then $\dots$ ”) is not possible. The absence of LEM also entails that the axiom of choice is not available (Diaconescu’s Theorem). But in our equational proofs we allow every concept that can easily be reduced to a “pure” equational proof. For rings, this includes the arithmetic of integers, modulo arithmetic, and calculations with polynomials. The statement “ x is central” abbreviates $xy = yx$ for a (free) variable y .

Therefore, equational proofs can be seen as the most direct and elementary proofs possible, even though they can be tough to find.

Example 2.4. Every boolean ring is commutative by an equational proof. We start with

$$-x = (-x)^2 = x^2 = x.$$

Then we calculate

$$x + y = (x + y)^2 = x^2 + xy + yx + y^2 = x + xy + yx + y$$

and hence $xy = -yx = yx$.

In contrast, the following proof of the same fact is *not* equational: Every reduced ring is a subdirect product of rings without zero-divisors [Kl80]. The only boolean rings without zero divisors are 0 and $\mathbb{F}_2$, which are commutative. So the subdirect product must be commutative as well. (Incidentally, this approach reduces Jacobson's theorem to the case of division rings.)

The second proof shows that every *model* of the theory of boolean rings is commutative. That this implies the existence of an equational proof is the content of Birkhoff's Completeness Theorem [Bi35]. But as mentioned in the introduction, this approach does not allow us to *find* an equational proof. From a constructive point of view, it is pretty much useless.

Definition 2.5. Let A be a rng.

- (1) We call A *potent* when for every $x \in A$ there is some natural number $n > 1$ with $x^n = x$.
- (2) Let $n > 1$. We call A an *n-rng* (or *n-ring* in the unital case) when $x^n = x$ holds for every $x \in A$.

Remark 2.6.

- (1) The class of n -rings has been studied for example in [MM37], [Sch10, Appendix B] (under the name *n-boolean rings*) and [Pi67, Section I.12]. If $p \in \mathbb{P}$, some authors require $pA = 0$ in the definition of a p -ring, but we will not do that.
- (2) There is an algebraic theory $\mathcal{T}_n$ whose models are n -rings: just add the axiom $x^n = x$ to the ring axioms. In particular, Definition 2.1 applies and we can speak of equational proofs in the theory of n -rings.
- (3) The definition of an n -ring only uses the multiplicative structure, so that *n-monoids* can be defined in the obvious way. But they are not automatically commutative. The smallest counterexample is the unitalization of the semigroup $\{a, b\}$ with $x \cdot y := x$.
- (4) Recently, Oman [Om23] has shown that a rng is potent if and only if every non-zero subrng contains a non-zero idempotent.
- (5) Jacobson's Theorem states that every potent rng, and hence every n -rng, is commutative [Ja45]. We refer to [Wa71, NT74, Do76] for elementary (albeit non-equational) proofs. We will see in a moment that it is actually sufficient to consider the unital case.

Remark 2.7. Let us gather some basic observations on potent rings and n -rings.

- (1) For every prime power q the finite field $\mathbb{F}_q$ is a q -ring. This follows from Lagrange's Theorem applied to $\mathbb{F}_q^\times$.
- (2) Every potent rng is reduced. This is because $x^2 = 0 \implies x = 0$ holds.
- (3) In particular, there is an equational proof that every n -rng is reduced (meaning that for every term τ an equational proof of $\tau^2 = 0$ leads to an equational proof of $\tau = 0$).
- (4) The class of n -rings is closed under product rings, quotient rings and subrings.
- (5) The class of potent rings is closed under finite products, but not under infinite products. For instance, $\prod_{p \in \mathbb{P}} \mathbb{F}_p$ is not a potent ring. One can even show that $(\mathbb{F}_p)_{p \in \mathbb{P}}$ has no product in the category of potent rings, which also shows that potent rings are not modelled by an algebraic theory.
- (6) Every potent ring that is an integral domain is a field. It follows that in a commutative potent ring every prime ideal is maximal, i.e. it has Krull dimension ≤ 0 . (Recall that the zero ring has Krull dimension $-\infty$.)
- (7) When n is even, there is an equational proof that every n -rng satisfies $-x = x$.

Lemma 2.8. *In a reduced rng, idempotents are central. In an n -rng, an equational proof of this claim is available.*